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ENBC 332

Professor Azarnoosh

11/22/2025

Homework 8 Discussion and Outputs

Problem 1

Part B:

The coefficient for the treatment of drug represents the estimated effect of the drug compared to the placebo. More specifically, its value of –4.33 suggests a 4.33 day on average shorter recovery time for patients that have received the drug compared to those who did not. The coefficient for the gender dummy represents the estimates whether male patients have a faster recovery compared to female patients when using the drug. The value of –0.6667 obtained indicates that the male patient is predicted to have 0.67 days shorter recovery on average compared to a female patient.

Part C:

General equation

$$Recovery\,Time = 13 - 4.33 \cdot (DrugGiven) - 0.67 \cdot (GenderMale)$$

Female with Drug:

$$13 - 4.33 \cdot 1 - 0.67 \cdot 0 \approx 8.67\,Days$$

Male with Drug

$$13 - 4.33 \cdot 1 - 0.67 \cdot 1 \approx 8\,Days$$

Problem 2

Part B:

Based on the significance level of 5%, the only impactful variable is the distance the house is from the city center. The p-value generated for this variable was 0.0267 which is below the significance level of 0.05, implying statistical significance. This is not the case for all of the other variables where their p-values are higher than the significance level.

Part C:

General Equation:

$$House\ Cost = 305.16 + 0.00218 \cdot (Size) + 2.65 \cdot (Bedrooms) + 0.31 \cdot (Age) - 1.807 \cdot (dist\ to\ city\ center)$$

Prediction Based on Given Values:

$$305.16 + 0.00218 \cdot (2500) + 2.65 \cdot (4) + 0.31 \cdot (20) - 1.807 \cdot (10) \approx 309.34$$

As a result, the predicted house price for the following parameters is approximately $309,340 dollars.

```
> # Christopher A. Lee
> # ENBC 332
> # Homework 9 Problem 1
> # 11/22/2025
>
> # Given Dataset
> Recovery_Time = c(10, 8, 15, 12, 7, 14, 9, 11, 6, 13)
> Treatment = c("Placebo", "Drug", "Placebo", "Drug", "Drug", "Placebo", "Drug", "Placebo", "Drug", "Placebo")
> Gender = c("Male", "Female", "Female", "Male", "Male", "Female", "Female", "Male", "Female", "Male")
>
> # Part A
> df_drug <- data.frame(Recovery_Time, Treatment, Gender)
> df_drug$Treatment <- factor(df_drug$Treatment, levels=c("Placebo", "Drug"))
> df_drug$Gender <- factor(df_drug$Gender, levels=c("Female", "Male"))
>
> model_drug <- lm(Recovery_Time ~ Treatment + Gender, data=df_drug)
> summary(model_drug)

Call:
lm(formula = Recovery_Time ~ Treatment + Gender, data = df_drug)

Residuals:
    Min      1Q  Median      3Q     Max 
-2.6667 -1.2500 -0.1667  0.9167  4.0000 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)    13.0000     1.3663   9.515 2.97e-05 ***
TreatmentDrug  -4.3333     1.4907  -2.907   0.0228 *  
GenderMale     -0.6667     1.4907  -0.447   0.6682    
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 2.309 on 7 degrees of freedom
Multiple R-squared:  0.5475,	Adjusted R-squared:  0.4182 
F-statistic: 4.234 on 2 and 7 DF,  p-value: 0.06234

> # Christopher A. Lee
> # ENBC 332
> # Homework 9 Problem 2
> # 11/22/2025
>
> # Given Dataset
> set.seed(123) # For reproducibility
> Price = round(rnorm(50, mean = 300, sd = 50), 2)
> Size = round(runif(50, 800, 4000), 2)
> Bedrooms = sample(2:6, 50, replace = TRUE)
> Age = sample(1:50, 50, replace = TRUE)
> Distance_to_City_Center = round(runif(50, 1, 30), 2)
>
> # Part A
> df_house <- data.frame(Price, Size, Bedrooms, Age, Distance_to_City_Center)
>
> model_house <- lm(Price ~ Size + Bedrooms + Age + Distance_to_City_Center, data=df_house)
> summary(model_house)

Call:
lm(formula = Price ~ Size + Bedrooms + Age + Distance_to_City_Center, 
    data = df_house)

Residuals:
    Min      1Q  Median      3Q     Max 
-95.177 -28.197  -6.875  28.571  99.622 

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)    
(Intercept)             305.156791  33.566563   9.091 9.43e-12 ***
Size                      0.002718   0.007044   0.386   0.7014    
Bedrooms                  2.649592   4.804899   0.551   0.5841    
Age                       0.305003   0.554300   0.550   0.5849    
Distance_to_City_Center  -1.807368   0.789040  -2.291   0.0267 *  
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 45.33 on 45 degrees of freedom
Multiple R-squared:  0.1196,	Adjusted R-squared:  0.04134 
F-statistic: 1.528 on 4 and 45 DF,  p-value: 0.2101
```